

Fig. 2: Socket arrangement

between the plug terminal and the socket causing damage to both.

The switch that controls the socket may also get damaged by this spark if the socket was initially turned off when the SMPS was plugged in. This reduces the lifetime of both the socket switch and the SMPS plug.

The circuit given here avoids the

damage by stopping fast charging of the capacitor inside the SMPS device through a resistor.

After a delay of about one second provided by monostable IC NE555 (IC2), the capacitor inside the SMPS charges. The relay energises, closing the contact and shorting the series resistor. As the power to the SMPS is applied after some delay, the capacitor gets sufficient time to charge and thus sparking is avoided. For normal operation, first press the reset switch (S2) and then plug the SMPS to L and N terminals (refer Fig. 1).

CD4013 (IC3) is a dual D-type flip-

flop. Out of the two flip-flops, only one is used here. The flip-flop is clocked by the pulse provided by timer IC2, which is configured as a monostable multi-vibrator.

When the circuit is switched on using power switch S1, IC1 provides clock pulse to the flip-flop. The Q1 output of IC3 goes high and the relay-driver transistor conducts to energise the relay. The relay remains energised until it is reset by switch S2.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. The proposed socket arrangement is shown in Fig. 2. ●

Security Alarm

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Thwart any attempt of burglary in your house using this alarm circuit. When someone opens the door of your room, it sounds an alarm intermittently and flashes light as well. The circuit can also be used as an audio/visual alarm in case of fire or other emergency by momentarily pressing switch S3.

The circuit (refer Fig. 1) is built around transformer X1, a standard bar magnet, reed switch S2, timer IC NE555 (IC1), opto-coupler IC MOC3020 (IC2), TRIAC BT136 and a few discrete components. Timer IC1

is wired as an astable multi-vibrator whose reset pin 4 is controlled by the reed switch. The reed switch fitted in the door frame acts as the sensor. A magnet is fixed on the door panel close to the reed switch.

The reed switch consists of a pair of contacts on ferrous metal sealed in a glass envelope. The contacts may be normally-open (which close when a magnetic field is present) or normally-closed type (which open when a magnetic field is applied). A normally-open-type reed switch is used here.

When the door is closed, reed switch S2 is in open state. When the

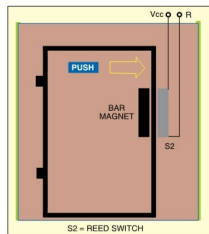


Fig. 2: Reed switch fitting in door

door is opened, the bar magnet moves away from reed switch S2. As a result, reset pin 4 of IC NE555 goes high. The high output at pin 3 of IC1 enables IC2. Pin 4 of IC2 is connected to the gate of TRIAC1.

When the door is opened, bulb B1 flashes and the bell sounds (provided switch S4 is closed) indicating that the door has been opened. Flashing of the bulb and the alarm continue until the door is closed.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Connect the call bell at the back side and the bulb at the front side of the cabinet. Install the unit on the door of the room as shown in Fig. 2.

The circuit is powered by mains supply. ●

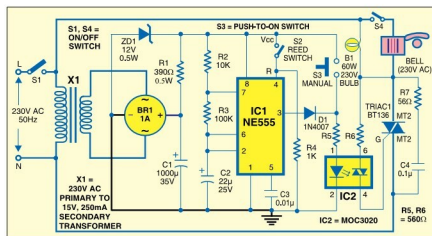


Fig. 1: Security alarm circuit